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Melting in High-Pressure Ice Layers of Large Ocean Worlds—Implications for Volatiles Transport

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Key Points:
- Water is produced at the silicates interface after differentiation and transported to the ocean with volatile material dissolved in it
- Thermal boundary layer analysis predicts the critical Rayleigh number at which bottom melting stops as the moon cools and the ice thickens
- Scaling laws for the thermal structure of the high-pressure ice layer and the outflow water velocity are derived from numerical modeling

Supporting Information:
- Supporting Information S1

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Abstract

A high-pressure ice layer controls the exchange of heat and material between the silicate core and the ocean of Ganymede and Titan. We have shown (Kalousová et al., 2018, http://doi.org/10.1016/j.icarus.2017.07.018) that a temperate (partially molten) layer is always present at the ocean interface. Another temperate layer with a few percent of water may be present at the silicates interface for low values of Rayleigh number. We derive scaling laws to predict the critical value under which this temperate layer exists and the amount of generated melt. The presence of liquid water in contact with silicates was probably limited to the early history, providing a pathway for the transfer of salts and volatiles like ^{40}Ar to the ocean. We also derive scaling laws for the water outflow velocity and for the top temperate layer thickness. These laws can be used to model the global thermal and compositional evolution of large ocean worlds.

Plain Language Summary

Ocean worlds, where a deep global ocean is present below the icy crust, provide an interesting habitable environment where life may exist. On Enceladus, which is small, and Europa, where the H_2O/silicate (water/rock) ratio is small, the global ocean is in direct contact with the silicates. On Titan and Ganymede, where this ratio is large, a layer of high-pressure (HP) ice is present between the ocean and the rocky core. This paper shows that early in their evolution, the lower part of this HP ice layer was temperate (porous ice with water in the pores). Such a temperate layer enables a silicates-ocean exchange of salts and volatiles such as ^{40}Ar that was measured in Titan's atmosphere by the Cassini mission. It also provides a potentially habitable environment in Ganymede, the largest moon in the solar system that will be studied by the ESA JUICE mission.

1. Introduction

Titan and Ganymede have a thick H_2O layer composed of a deep salty water ocean sandwiched between an outer ice I crust and an inner high-pressure (HP) ice layer (e.g., Hussmann et al., 2015). This structure and the salty nature of the ocean is inferred from gravity (Iess et al., 2010, 2012; Mitri et al., 2014) and electromagnetic data (Beghin et al., 2012; Kivelson et al., 2002) obtained by the Galileo and Cassini missions. Below the HP ice layer is an interior composed of hydrated (Titan) or dehydrated (Ganymede) silicates (Castillo-Rogez & Lunine, 2010; Scott et al., 2002). The particular type of salts in the ocean is unknown, but magnesium sulfate has been proposed for Ganymede (Vance et al., 2014) and ammonium sulfate for Titan (Fortes et al., 2007). Other components such as carbonates and sodium and potassium chlorides may also be present (Mitri et al., 2014). These ions were formed by leaching of silicates either during or after the differentiation and had to be transported into the ocean. In addition, the large amount of ^{40}Ar in Titan's atmosphere as well as the presence of CH_4 which should disappear in a few tens of million years (Yung et al., 1984) provide the evidence of exchange between the interior and the surface.

Understanding the transfer of heat and mass through the HP ice layers is crucial for understanding the habitability of these ocean worlds (e.g., Nimmo & Pappalardo, 2016). One important aspect is whether liquids can form at the silicate/HP ice interface. In our previous study (Kalousová et al., 2018), we have demonstrated that ice melts at this interface if the Rayleigh number (Ra) that characterizes the vigor of convection is small. For large values of Ra, convection becomes efficient enough that the temperature at the silicates interface becomes lower than the melting temperature. As the moon cools, its HP ice layer thickens leading to larger

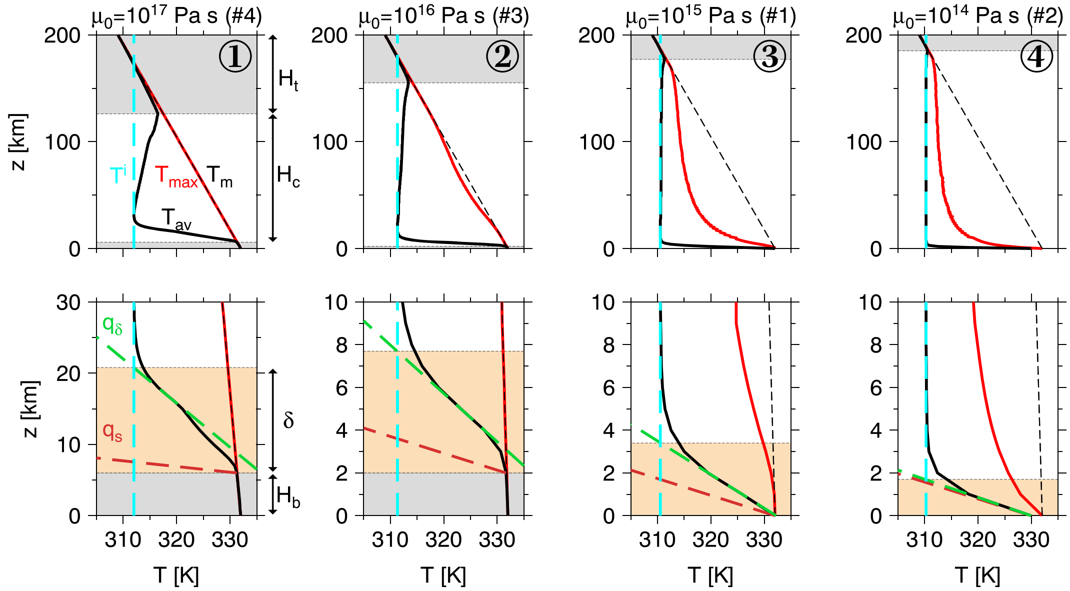

Figure 1. Temperature profiles of each of the four regimes (circled numbers) computed with H = 200 km, the prescribed silicates heat flux q_s = 20 mW/m^2, and the reference viscosity μ_0 decreasing from left to right (simulation # corresponds to Table S1). (top) Maximum (red), horizontally averaged (black), melting (black dashed), and interior (cf. text, cyan dashed) temperature, top (H_t) and bottom (H_b) temperate layers (gray rectangles). (bottom) Detail of silicates interface—hot TBL (δ, brown rectangles), prescribed silicates heat flux (q_s, brown dashed), and computed TBL heat flux (q_δ, green dashed).

Ra values, which suggests that transport of volatiles from the silicates to the ocean is limited to the early periods of the moon's evolution. If water is generated at the silicates interface, it is transported very efficiently to the ocean by upwelling plumes. In the present study, we investigate in more detail the boundary layers at the interface with the silicates and the deep ocean and we provide the implications for the transfer of volatiles from the interior to the ocean.

2. Results

We treat the HP ice layer as a two-phase mixture of solid ice and liquid water with no impurities. The governing equations comprise the mass and linear momentum balances of both phases as well as the energy balance of the whole mixture which is considered to be in thermal equilibrium (Souček et al., 2014). The numerical model is implemented in the software package FEniCS (Alnaes et al., 2015; Logg et al., 2012) and described in more detail in the supporting information (section S1). Resolution tests were performed which showed that the steady state temperature fields were well resolved on the chosen mesh. We use the results of numerical simulations from Kalousová et al. (2018) and of new simulations designed specifically for this study. Table S1 in the supporting information summarizes all simulations and their outputs. Our results show that the HP ice layer of thickness H can be divided into three parts: a layer of temperate ice (i.e., ice where temperature equals the melting temperature) at the interface with the ocean—we will denote its thickness H_t, a convective layer of thickness H_c, and the bottom boundary where another temperate layer may be present—its thickness being H_b. Clearly, $H = H_b + H_c + H_t$.

We investigate the possibility of volatiles exchange between the silicates and the liquid ocean, which can be significantly facilitated by the presence of water at the silicates interface since volatiles can be dissolved in water and then extracted into the ocean. Such a process is more efficient than if volatiles would have to be captured and advected by the convecting solid ice. In Kalousová et al. (2018) we identified three regimes with different degree of material exchange between the silicates and the liquid ocean. Here after a more detailed analysis, we divide the intermediate regime into two based on the presence or not of a temperate layer at the interface with the silicates. The resulting division in terms of the degree of exchange between the silicates and the ocean is as follows (with increasing Ra):

1. *Direct exchange.* The temperature at the silicates interface equals the melting temperature ($T^s = T^s_m$) and a bottom temperate layer is present where melting occurs. The generated water is advected through the

convecting interior by upwelling plumes of temperate ice that are stable in time and space, thus assuring that water does not freeze during the ascent. More melting may occur during the ascent and within the top temperate layer at the ocean interface (Figure 1, first column).

2. *Indirect exchange.* The temperature at the silicates interface equals the melting temperature and a bottom temperate layer is present. However, most of the water generated at this interface freezes during the ascent through the convecting interior. Then melting occurs again in the top temperate layer (Figure 1, second column).
3. *Limited exchange.* The temperature at the silicates interface equals the melting temperature, but no temperate layer is present ($H_b \ll h$, with h the grid element). Only a limited amount of water is generated and most of it freezes during the ascent. Melting occurs again in the top temperate layer (Figure 1, third column).
4. *No exchange.* The heat transfer by solid state thermal convection is so efficient that the average temperature at the silicates interface is below the melting point ($T^s < T^s_m$) and no water is produced (Figure 1, fourth column). Melting however occurs in the top temperate layer.

Let us note that even though water is generated at the silicates interface in three out of four regimes, no significant amount is accumulated at this interface.

2.1. Melting at the Silicates Interface—Thermal Boundary Layer Analysis

Melting does not occur at the silicates interface if the temperature T^s is below the melting temperature T^s_m which happens if convection is vigorous enough. The temperature contrast across the hot thermal boundary layer (TBL) can be written as

$$\Delta T_c = T^s - T^i < T^s_m - T^i \tag{1}$$

where T^i is the temperature of the convective interior (called interior temperature henceforth, cyan lines in Figure 1). For simulations in regime 4, it well corresponds with the melting temperature at the base of the top temperate layer, T^t_m (Figure 1 right), which can be evaluated if its thickness H_t is known (section 2.3). In this regime, convection is driven by hot instabilities that form in the hot TBL at the silicates interface. There is no cold TBL at the ocean interface as the average temperature profile follows the melting curve. HP ice convection is thus an example of convection with only a hot TBL, contrary to volumetrically heated fluids (Rayleigh-Roberts heating configuration) which have only a cold TBL or to Rayleigh-Bénard convection which has both hot and cold TBLs. We employ the far-field Rayleigh number Ra^+ defined as (e.g., Sotin & Labrosse, 1999):

$$\mathrm{Ra}^+ = \frac{\alpha_i \rho_i^2 g\, c_i^p \Delta T_c H_c^3}{k_i \mu_0} \tag{2}$$

(material parameters are listed in section S1). The advantage of using Ra^+ instead of Roberts Ra_R or Bénard Ra_B numbers is that it is defined independently of the heating type by using the temperature difference across the TBL (ΔT_c) and the convective layer thickness (H_c) and thus allows comparison between the configurations with internal heating, heating from below (prescribed heat flux), and constant temperature boundary conditions. Let us also note that even though we are using temperature and pressure dependence of ice viscosity (equation (S3)), its variations in the convective layer are small ($\mu_{max}/\mu_{min} < 10$ for the thickest layers). The viscous temperature scale (Davaille & Jaupart, 1993) is several times larger than the temperature difference across the TBL and thus performing the analysis for isoviscous material is more appropriate here.

The thermal boundary layer analysis uses the relationship between the nondimensional TBL thickness δ' defined as

$$\delta' = \frac{\delta}{H_c} \tag{3}$$

(with δ the dimensional TBL thickness, cf. Figure 1) and the far-field Rayleigh number Ra^+ with the particular form of scaling law depending on the boundary conditions. The scaling for Rayleigh-Bénard convection with free slip (Jarvis, 1984) and no slip (Mitrovica & Jarvis, 1987) boundaries is discussed in the supporting information. In the present study, the boundary conditions are no slip at the bottom and free slip at the top boundary. Moreover, heat flux from silicates q_s is prescribed at the bottom boundary (instead of a fixed temperature for Rayleigh-Bénard convection). For simulations in regime 4 (no bottom melting), we found the following scaling (Figure S1, left):

$$\delta' = 2.746(\mathrm{Ra}^+)^{-0.271} \tag{4}$$

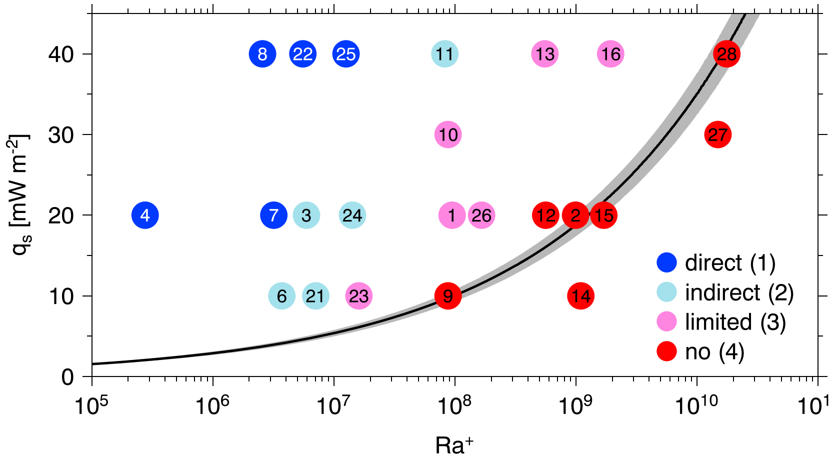

Figure 2. Values of (Ra$^+$, q_s) for all simulations (numbers correspond to Table S1) sorted by exchange regime (colors). The scaling law defined by equation (7) (black line) and its range for $H = 100-400$ km (gray, equation S15) predicts the transition from the presence of melt in contact with the silicates (regimes 1–3) to no melt at the silicates interface (regime 4).

With this expression for the TBL thickness, we can determine the critical Ra$^+$ above which there is no melting. Using the fact that if there is no melting, then

$$q_s = q_\delta = k_i \Delta T_c / \delta \tag{5}$$

(cf. green and brown lines in the right bottom panel in Figure 1), substituting equations (3)–(5) into (1), and assuming that the critical value is such that the inequality changes to equality, we can write

$$\mathrm{Ra}_c^+ = \left(2.746 \frac{q_s H_c}{k_i (T_m^s - T^i)} \right)^{3.690} \tag{6}$$

Using the scaling for H_t and T^i (equation (9) and section S3), we evaluated Ra_c^+ for a range of input parameters (H, μ_0, and q_s), found that it strongly depends on the silicates heat flux q_s (Figure S1, right) and not so much on $H_c/(T_m^s - T^i)$ because the larger H_c, the larger $T_m^s - T^i$. We then found the following scaling:

$$\mathrm{Ra}_c^+ = 19.965 \times 10^3 (q_s[\mathrm{mW/m^2}])^{3.690} \tag{7}$$

Figure 2 shows the values of (Ra$^+$, q_s) for all simulations. The red circles correspond to simulations in regime 4, the black line to the scaling defined by equation (7), and the gray region shows the scaling law range for H=100–400 km (equation (S15)). All red circles except for #12 lie on the scaling law line or to the right, indicating that the scaling gives a good estimate for the presence or not of bottom melting. In the case of simulation #12, the temperature at the bottom boundary is very close to melting temperature ($|T^s - T_m^s| < 1$ K) indicating that this simulation is not far from being in regime 3 (cf. also the fourth panel in Figure 8 in Kalousová et al., 2018, which shows that indeed $T^s \lesssim T_m^s$ for #12).

Using this scaling, we can compare the value of Ra_c^+ with that of the far-field Rayleigh number, which depends on the characteristics of the HP ice layer, including the thickness of the convective layer H_c and the interior temperature T^i. If the far-field Rayleigh number is larger than critical, then no water is generated at the silicates interface. While H_c and T^i are the results of numerical simulations, scaling can be found for H_t as a function of input parameters (section 2.3), and the interior temperature can be approximated by melting temperature at the base of the top temperate layer: $T^i \sim T_m(z = H - H_t)$. Since the thickness of the bottom temperate layer H_b (if present) is small compared to H (Figure S2, left), it can be neglected in this analysis and thus $H_c = H - H_t$.

The porosity at the silicates interface is always only a few percent. Let us define the equivalent heat flux consumed on melting within the first layer of elements:

$$Q_m^0 = \frac{L}{2H} \int_0^{2H} \int_0^{z_0} r \mathrm{d}x \mathrm{d}z \tag{8}$$

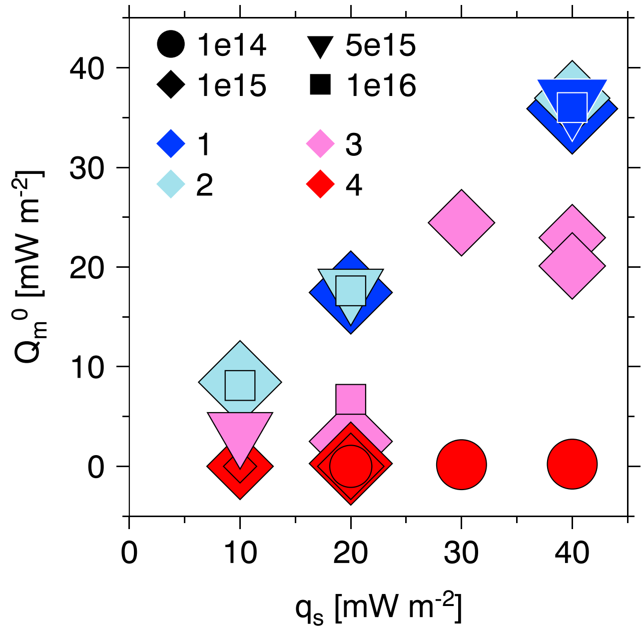

Figure 3. Heat consumed on bottom melting Q_m^0 as a function of silicates heat flux q_s sorted by exchange regime (colors) and viscosity (symbols). In regimes 1 and 2, where the bottom temperate layer is present, all the heat coming from silicates is consumed on melting: $Q_m^0 = q_s$.

where r is the melting rate and z_0 is the height of the bottom element. Figure 3 shows Q_m^0 as a function of the silicates heat flux q_s and indicates that in regimes 1 and 2 ($H_b > 0$, dark and light blue) all the supplied energy is consumed on melting in the bottom temperate layer, and thus, the melting rate is controlled by the silicates heat flux: $r \sim 4\pi R_s^2 q_s / L$. For q_s between 10 and 40 mW/m^2, $R_s = 2{,}080$ km (Titan case, Castillo-Rogez & Lunine, 2010), and $L = 306$ kJ/mol, the melting rate is between 5.6×10^{13} to 2.2×10^{14} kg/year. We provide some implications for the leaching of argon in section 3.

2.2. Transfer Within the Convective Layer

As mentioned above, no cold TBL can form at the top boundary with the ocean, and convection is thus characterized by hot upwelling plumes that form in the hot TBL at the interface with the silicates.

To estimate the time necessary for the leached material to be transferred from the silicates to the ocean, we investigate the velocities. In regimes 2 and 3, the generated water (and thus the volatile material dissolved in it) freezes during the ascent and so the advection velocity is that of the convecting ice. In regime 1, water stays in the liquid state but its amount is small and scarcely exceeds the percolation threshold ϕ_c (Golden et al., 1998), which makes the advection velocity again very close to that of convecting ice. The root-mean-square (rms) velocities of ice and water differ by less than 1% for regime 4 and (with the exception of simulation #4) by 1–15% for regimes 1–3 (Figure S3, left). In general, the ice and water velocities only differ significantly in the top temperate layer. The rms ice velocities for simulations in regimes 1–3 are within the range of 0.1–1 m/year (Figure S3, right) giving the travel times from the silicates to the ocean on the order of hundreds of thousands of years to few millions of years.

2.3. Top Temperate Layer

A layer of temperate ice always forms at the top interface with the ocean, its thickness depending on the viscosity μ_0 and heat flux q_s (Figure S2, right). Combining these two dependencies, we found the following scaling law (cf. section S3 for more details):

$$H_t[\text{km}] = (0.145 \times 10^{-3} q_s[\text{mW/m}^2] + 0.015)\mu_0[\text{Pa} \cdot \text{s}]^{0.21} \tag{9}$$

Left panel of Figure 4 shows the thickness H_t evaluated using the scaling law (equation (9)) as a function of the thickness H_t that we obtained from numerical simulations and indicates that equation (9) provides a very good estimate of the observed thickness H_t. The rms error is 2.4 km.

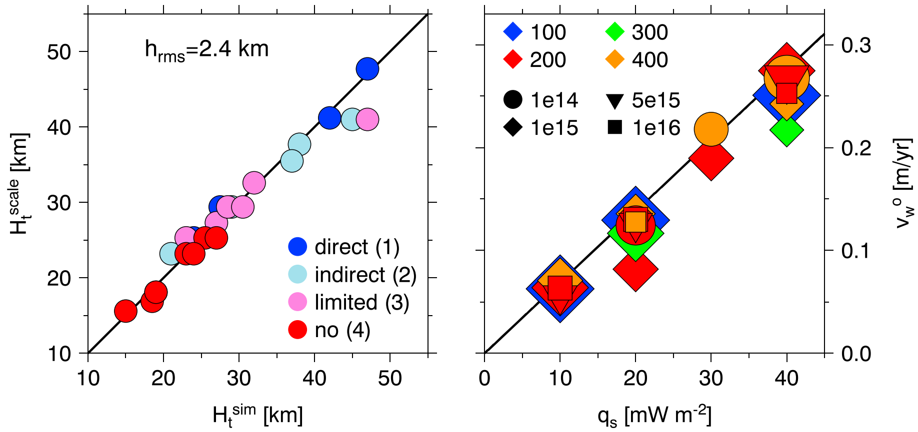

Figure 4. (left) Top temperate layer thickness H_t: scaling (equation (9)) versus values from simulations sorted by exchange regime (colors). The rms error is 2.4 km. (right) Outflow water velocity v_w^o as a function of silicates heat flux q_s sorted by layer thickness (colors) and viscosity (symbols). Line indicates scaling (equation (10)). This plot shows a very good agreement between the results of numerical simulations and scaling laws.

Once steady state is achieved, the top temperate layer thickness does not vary and the average porosity within is given by the percolation threshold ϕ_c. Therefore, any extra water that comes from melting of ice has to flow out into the ocean and the average outflow velocity is also constant. Because almost all heat within the temperate ice is consumed on melting (heat flux along the melting curve is several orders of magnitude smaller), the amount of molten water can be estimated by $q_s S/L$ (with S the area of the interface with the ocean), while the water flux out of the layer is $\phi_c v_w^o \rho_w S$. Therefore, the average (in time and space) outflow velocity can be estimated by

$$v_w^o = \frac{q_s}{\phi_c \, \rho_w \, L} \tag{10}$$

Right panel in Figure 4 shows the outflow water velocity v_w^o as a function of prescribed silicates heat flux q_s. Colors indicate the HP ice layer thickness H and shapes the melting point viscosity μ_0. The line corresponds to equation (10) evaluated with $\rho_w = 1{,}270$ kg/m^3 and shows a very good agreement with the observed values. Equation (10) suggests that other parameters (viscosity, HP ice layer thickness) have negligible effect on the outflow water velocity, while the increasing percolation threshold decreases the outflow water velocity which is supported by our results (not shown here). Let us note that the outflow velocities vary in space and time and their maxima are well correlated with the hot ice upwellings (Figure S4). Also note that melting occurs as the hot upwelling plumes reach the temperate layer, and thus, the heat that was transferred by convection through the interior is in the top temperate layer transferred by melting of ice and outflow of water into the ocean.

3. Discussion

One objective of this paper is to provide a scaling law which can be implemented into one-dimensional evolution models of Titan and Ganymede in order to predict whether melting will occur at the silicates interface and to estimate the potential for exchange processes between the silicate core and the deep ocean. Such models are important for the interpretation of Cassini data (Titan) and for preparing JUICE observations (Ganymede). The model parameters include the melting curve of HP ice (well known) and the ice viscosity at the melting point μ_0 that is known within 2 or 3 orders of magnitude (Durham et al., 1996; Sotin et al., 1985; cf. also synthesis in Kalousová et al., 2018). One-dimensional evolution models provide the heat flux coming from the silicate layer q_s and the freezing rate of the HP ice layer (and thus its thickness H). Using the scaling for the top temperate layer thickness (equation (9)) and by approximating the interior temperature as $T^i \sim T_m(z = H - H_t)$, we can compute the value of the far-field Rayleigh number (assuming that $H_b = 0$). Comparison with the critical value for a given heat flux (equation (7)) allows us to determine whether water can be generated at the silicates interface and thus assess the potential for the transport of volatiles from the silicates to the ocean. One-dimensional evolution models using this scaling will place bounds on the timing of exchange processes and their results can be compared with the amount of ^{40}Ar delivered into Titan's atmosphere.

^{40}Ar is produced by decay of ^{40}K in the silicate core. Assuming that all argon diffuses to the surface of the rocky core (a hypothesis used for terrestrial planets, e.g., Kaula, 1999, for Venus), we can calculate its flux. Using Titan's mass and MoI to estimate the amount of silicates, assuming that the core is made of antigorite (Castillo-Rogez & Lunine, 2010), and employing terrestrial ^{40}K/Si ratio, we arrive at the initial argon flux of 4.6×10^7 mol/year, which slowly decreases with time (half life of ^{40}K is 1.277 Gyr). The flux of water that is produced and then transferred to the ocean is between 5.6×10^{13} and 2.2×10^{14} kg/year (section 2.1). With the flux of argon given above, the necessary solubility is between 0.2 and 0.8 μmol/kg H_2O — much less than the solubility of argon in seawater which is about 15 μmol/kg (and varies with temperature and salinity, cf. Hamme & Emerson, 2004). It is therefore likely that all available argon can be dissolved and then transported into the ocean.

Can water outflow from the HP ice temperate layer influence circulation in the deep ocean? Modeling convection in the deep ocean of Europa, Soderlund et al. (2014) found mean radial current speeds of 3 cm/s and zonal speeds typically 250 cm/s. For comparison, ocean currents in the Gulf Stream can exceed 1 m/s (although driven by winds rather than convection). Convection processes in the oceans of Ganymede and Titan should have slightly lower velocities because their spin rates (Coriolis forces) are twice and 4 times smaller, respectively. Our numerical simulations predict that the outflow velocity of water from the HP ice is at most $\sim$0.3 m/year, which is 6 orders of magnitude smaller than the radial speed due to convection processes in the ocean. The water outflow from the HP ice should therefore not influence the flow in the ocean.

4. Concluding Remarks

Using the results of two-dimensional simulations of two-phase convection in the HP ice layer of a large icy moon, we investigated the characteristics of the boundary layers at the interface with the silicates and the deep ocean. We found a scaling law that gives the value of critical Rayleigh number above which no water is generated at the interface with the silicates. This value strongly depends on the heat supplied from the silicate interior. Below the critical value, melting rate is governed by the heat flux coming from silicates. Comparison of the estimated amounts of water and argon indicates that all available argon can be transported into the ocean. These processes are more likely to happen at the beginning of the thermal evolution — as the moon cools in time, the thickness of its HP ice layer increases, and so does the Rayleigh number making the HP ice layer less permeable.

The generated water is transported through the convecting interior (either liquid of frozen) on time scales of 10^{-1} – 10^{0} Myr and reaches the top temperate layer before being extracted into the ocean. We show that the thickness of this layer depends on the ice viscosity and the heat flux from silicates, while the outflow water velocity (negligible when compared with the convective velocities in the ocean) depends on the silicates heat flux and the percolation threshold. These scaling laws can be used in the models of global thermal and compositional evolution of Ganymede and Titan as well as extrasolar ocean worlds. Such models will provide estimates of the timing of exchange processes and, in case of Titan, their results can be compared with the amount of ^{40}Ar in the atmosphere.

Finally, the determination of Ganymede's ice crust thickness by the JUICE mission will constrain the thickness of the HP ice thickness (Vance et al., 2014). The results of this study will then predict whether water is generated at the silicates interface and whether volatiles from this interface are still being transported to the ocean. Because the presence of salts modifies the melting curve and the densities, we do not draw any conclusion about their transport here. Their effect will be investigated in future work.

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